Economic Integration Agreements and Agricultural Trade

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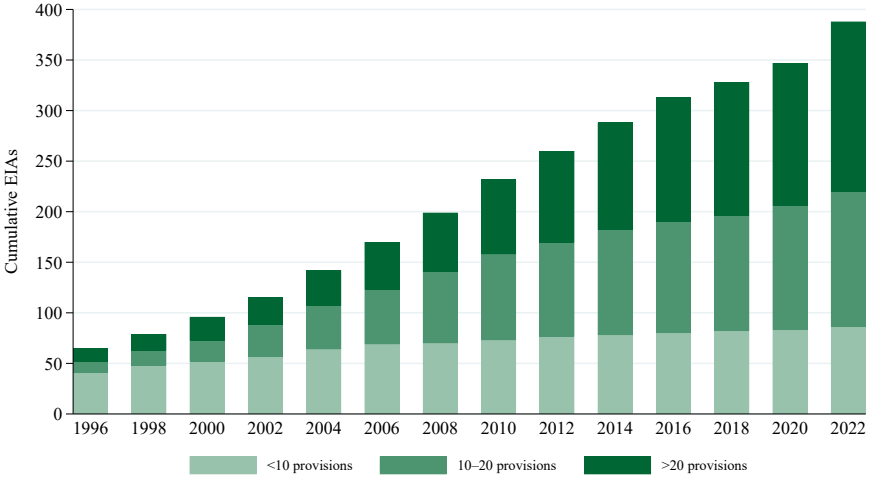
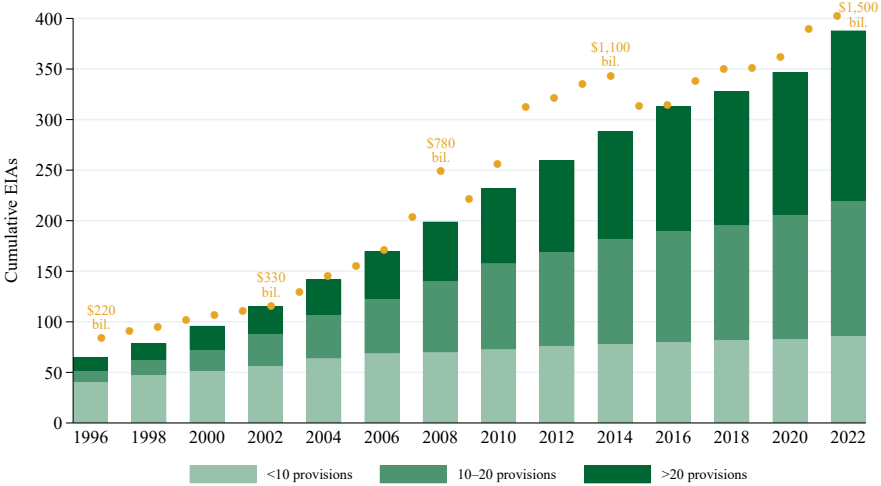
Economic Integration Agreements (EIAs) on the rise



And so are global agricultural exports



Understanding EIA effects on agricultural trade

- Growing accebility of granular data
  - trade flow decomposition
- Advances in trade theory
  - could link fixed costs to extensive margin
  - could understand heteregeneous responses across countries and products
  - could separate welfare channels (prices vs. variety effect)

Related Literature

EIAs are widely found to create trade in agricultural products: Grant & Lambert (2008), Lambert & Mckoy (2009), Sun & Reed (2010), Kim & Steinbach (2023).

→ This study contributes by decomposing trade into extensive and intensive margins to further explore where EIA-driven trade creation occurs at the product level.

Research shows inconsistent effects of EIAs on agricultural trade margins: Liapis (2009), Hejazi et al. (2017), Zhang et al. (2017), Sun & Li (2018), Scoppola et al. (2018).

→ First, this study provides the most comprehensive analysis of EIAs' effect on agricultural trade margins.

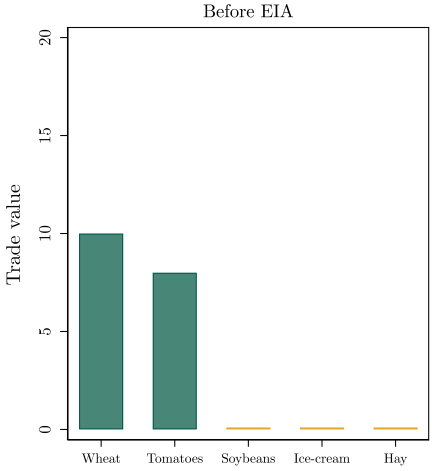
→ Second, it applies the 'new goods margin' defination to agricultural trade for the first time.

→ Third, it moves beyond average effects by exploring how and under what conditions EIAs affect trade margins across provisions, product types, country pairs, and over time.

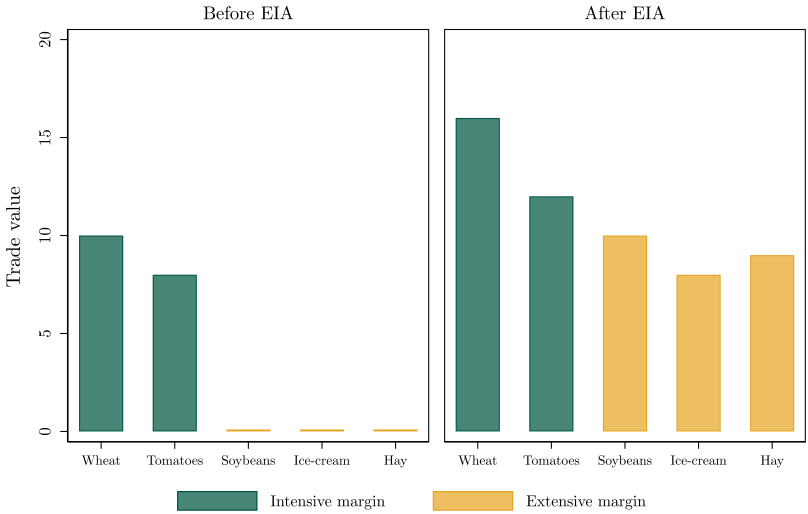
Research Questions

- The primary objective is to evaluate the impact of EIAs on the extensive (new trade flows) and intensive (pre-existing trade flows) margin of agricultural product trade.
- The following questions asked:
  - How does EIAs affect the agricultural trade through the margins?
  - How does EIA depth affect the agricultural trade and the margins?
  - How do EIA effects differ between raw and processed goods, and between low-income and high-income countries?
  - How do specific key provisions affect agricultural trade margins?

The two margins of export expansion



The two margins of export expansion



Measuring the extensive margin

- Following French & Zylkin (2024), this study defines the extensive margin, $\mathbf{1}_{k \in EX_{ij}^{10}}$, for each EIA country pair based on their pre-EIA bilateral trade volume:

$$s_{ijk}^{pre} = \frac{\bar{Y}_{ijk}^{pre}}{\bar{Y}_{ij}^{pre}} = \frac{\frac{1}{3}\sum_{s=t_0-6}^{t_0-4} Y_{ijks}}{\sum_{k \in K} \frac{1}{3}\sum_{s=t_0-6}^{t_0-4} Y_{ijks}} \tag{1}$$

- s_{ijk}^{pre} then ranked in ascending order and assign $\mathbf{1}_{k \in EX_{ij}^{10}} = 1$ to those in the bottom 10% of the distribution.
- The remaining goods, comprising the remaining 90% of total bilateral trade, form the intensive margin, $\mathbf{1}_{k \in EX_{ij}^{10}} = 0$
- The resulting margins are unique to each EIA country pair.
- The classification of goods into these margins is fixed in each pair's pre-agreement period.

Econometric approach

- The following pooled product-level gravity specification is estimated using PPML:

$$Y_{ijkt} = \exp\left(\beta_1 EIA_{ijt} + \beta_2\left(EIA_{ijt} \times 1_{k \in EX_{ij}^{10}}\right) + \beta_3 WTO_{ijt} + \mu_{ikt} + \gamma_{jkt} + \lambda_{ijk}\right)\varepsilon_{ijkt} \quad (2)$$

where $_i$ denotes the exporter, $_j$ the importer, $_k$ the product, and $_t$ the time.

μ_{ikt}, γ_{jkt}, and λ_{ijk} are Fixed effects.

$1_{k \in EX_{ij}^{10}}$ is equal to 1 if k belongs to the set of Extensive margin, EX_{ij}^{10}, between i and j.

ε_{ijkt} is the error term.

- β_1: Average partial effect of EIAs when $\mathbf{1}_{k \in EX_{ij}^{10}} = 0$.
- β_2: Differential partial effect of EIAs on $\mathbf{1}_{k \in EX_{ij}^{10}}$ relative to the base group.
- $\beta_1 + \beta_2$: Total average partial effect of EIAs when $\mathbf{1}_{k \in EX_{ij}^{10}} = 1$.

Data

Trade data

- Trade data come from BACI (CEPII) database. The sample includes 190 countries and 680 unique traded agricultural goods at HS96 6-digit level from 1996 to 2023. Agricultural goods are defined using USDA's BICO-HS6 classification.

Trade agreements data

- Agreements data come from the NSF-Kellogg Institute Database on EIAs (up to 2017). - DTA to update EIA through 2023. - DESTA to enrich it with provisions.
- EIA divides the agreements into six categories, depending on the depth:

Regular EIAs:
1. One-way PTAs
2. Two-way PTAs
3. FTAs

Deep EIAs:
4. Custom Unions
5. Common Markets
6. Economic Unions

WTO data

- WTO membership data come from the ITPD dataset, which covers up to 2019 and is extended to 2023.

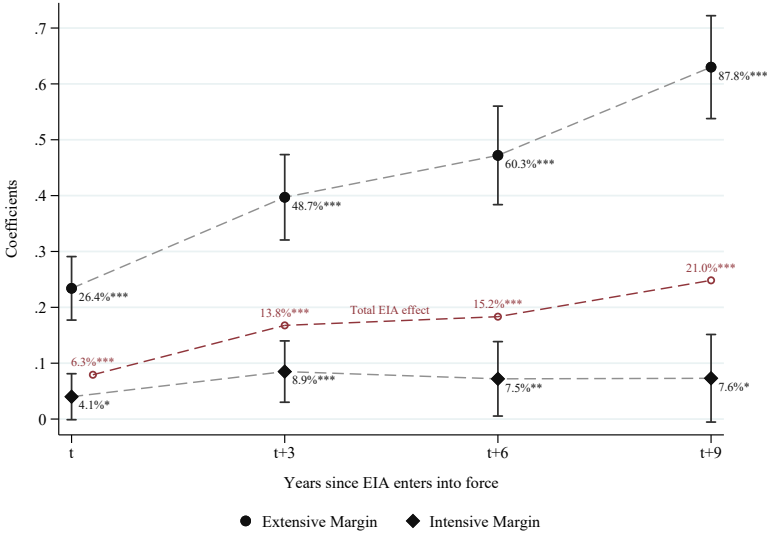
Finding 1: Extensive margin leads the way

Table: EIA Effects on Agricultural Exports and Margins

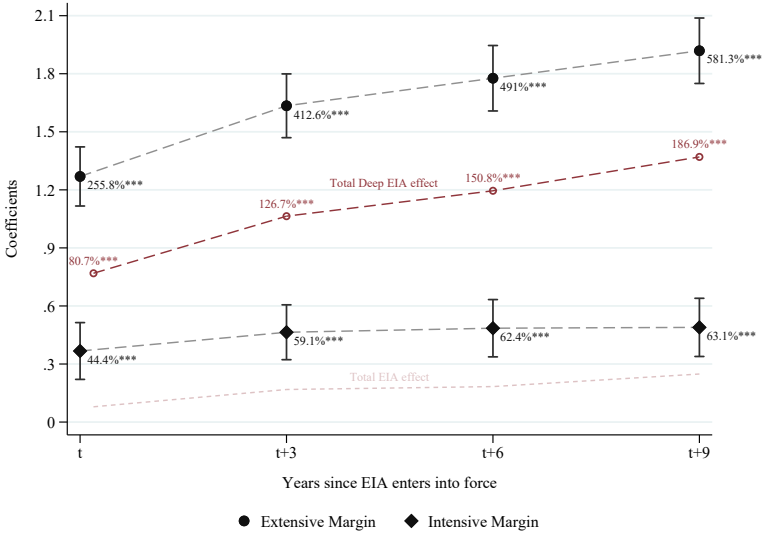
	(1)	(2)	(3)	(4)
All EIAs	0.117***		0.061**	
	(0.023)		(0.025)	
Shallow EIAs		0.127***		0.074***
		(0.023)		(0.025)
Deep EIAs		0.847***		0.431***
		(0.065)		(0.067)
All EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$			0.360***	
			(0.033)	
Shallow EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$				0.348***
				(0.032)
Deep EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$				1.289***
				(0.054)
WTO	0.138	0.160*	0.137	0.156*
	(0.091)	(0.090)	(0.091)	(0.090)

N = 19,293,038 Pseudo R^2 = 0.972 FEs = ikt, jkt, ijk SEs = ij

Finding 2: And it persists over time



Finding 3: The expansion deepens with depth

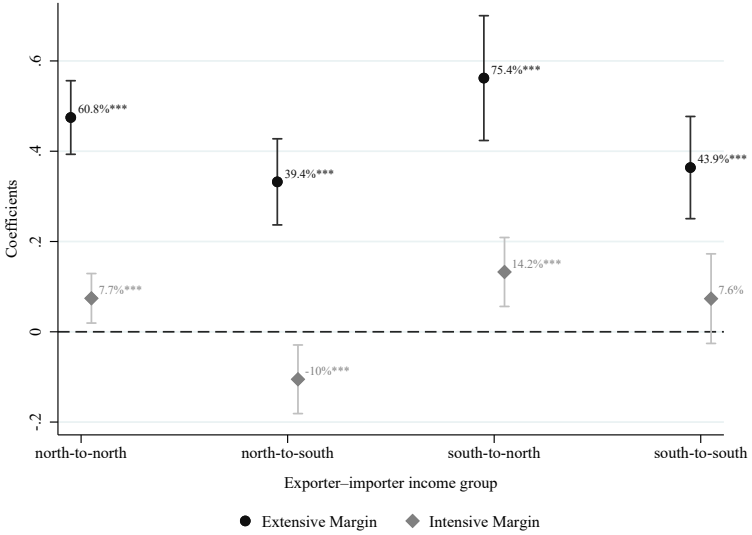


Finding 4: Value-added goods growth more

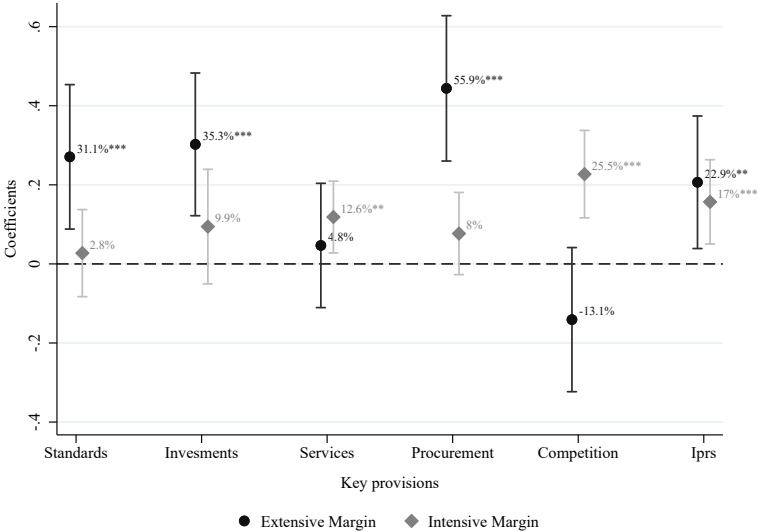
Table: EIA Effects by Processing Type

	Panel A. Processed Food Exports				Panel B. Raw Agricultural Exports			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
All EIAs	0.128***		0.066**		0.040		0.017	
	(0.028)		(0.030)		(0.056)		(0.058)	
Shallow EIAs		0.139***		0.079***		0.048		0.025
		(0.028)		(0.029)		(0.056)		(0.058)
Deep EIAs		0.872***		0.486***		0.906***		0.669***
		(0.065)		(0.064)		(0.133)		(0.141)
All EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$			0.420***				0.213**	
			(0.037)				(0.106)	
Shallow EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$				0.418***				0.214**
				(0.036)				(0.105)
Deep EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$				1.240***				0.852***
				(0.053)				(0.142)
WTO	-0.064	-0.034	-0.066	-0.038	0.380**	0.409**	0.377**	0.402**
	(0.084)	(0.083)	(0.084)	(0.082)	(0.182)	(0.181)	(0.183)	(0.182)
N	11,363,060	11,363,060	11,363,060	11,363,060	1,755,673	1,755,673	1,755,673	1,755,673
Pseudo R^2	0.972	0.972	0.972	0.972	0.970	0.970	0.970	0.970
FEs	ikt, jkt, ijk	ikt, jkt, ijk	ikt, jkt, ijk	ikt, jkt, ijk	ikt, jkt, ijk	ikt, jkt, ijk	ikt, jkt, ijk	ikt, jkt, ijk

Finding 5: Higher response when importers are high-income



Finding 6: Key provisions that drive the expansion



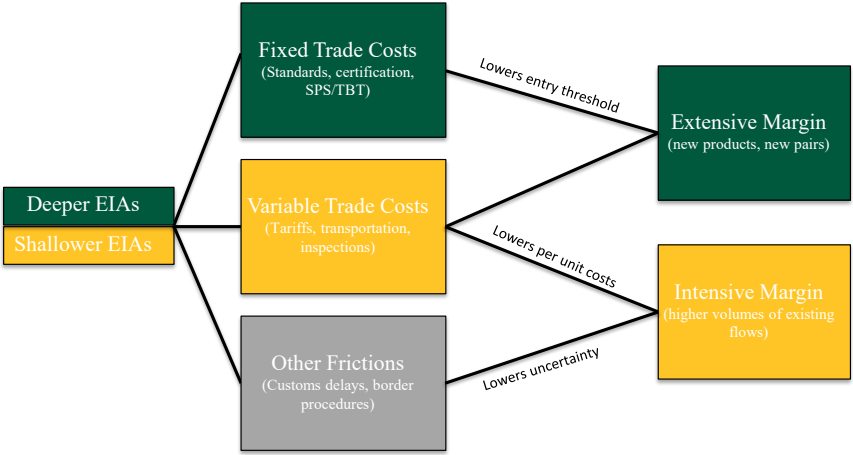
Conclusion

- The agreements, especially deeper ones, open the door to **new export opportunities**, particularly **for agri-goods that hadn't been traded intensely before.**
- **Value-added ag products grow faster** under trade deals, while raw goods see fewer gains, unless the agreement is deep.
- **Growth at the extensive margin** is especially strong **when developing countries export to developed markets**, and vice versa.
- The gains happen over time; **new agri-good exports** continue to **grow for at least 10 years** after the agreement starts.

Thank you!

Appendix

- Appendix 1: Main specification – lagged
- Appendix 2: Lagged specification results
- Appendix 3: Main specification – income group
- Appendix 4: Subsample income group results
- Appendix 5: Main specification – provisions
- Appendix 6: Variance inflation factor test
- Appendix 7: Naive regression
- Appendix 8: Predicted flows by EIA types
- Appendix 9: Lagged specification results by processing type
- Appendix 10: Lagged processed exports figure
- Appendix 11: Lagged raw exports figure
- Appendix 12: Further analyses
- Appendix 13: Aggregated at commodity group level – trimmed
- Appendix 14: Aggregated at commodity group level – full sample
- Appendix 15: Agregated results
- Appendix 16: Aggregation bias


Deeper EIAs
Shallower EIAs
Fixed Trade Costs
(Standards, certification, SPS/TBT)
Variable Trade Costs
(Tariffs, transportation, inspections)
Other Frictions
(Customs delays, border procedures)
Lowers entry threshold
Lowers per unit costs
Lowers uncertainty
Extensive Margin
(new products, new pairs)
Intensive Margin
(higher volumes of existing flows)

Appendix 1

$$Y_{ijkt} = \exp\Bigg(\sum_{\ell \in \{0,1,2,3\}} \Big[\beta_{1,\ell}\, EIA_{ij,t-\ell} + \beta_{2,\ell} \left(EIA_{ij,t-\ell} \times \mathbf{1}_{k \in EX_{ij}^{10}} \right) \Big] + \beta_3\, WTO_{ijt} + \mu_{ikt} + \gamma_{jkt} + \lambda_{ijk} \Bigg) \varepsilon_{ijkt} \tag{3}$$

Notes: $\ell \in \{0, 1, 2, 3\}$ indexes event-time lags corresponding to 0, 3, 6, and 9 years after implementation. $\mathbf{1}_{k \in EX_{ij}^{10}}$ denotes the extensive-margin indicator based on the 10% threshold. $\beta_{1,\ell}$ and $\beta_{2,\ell}$ capture dynamic EIA effects on the intensive and extensive margins, respectively. μ_{ikt}, γ_{jkt}, and λ_{ijk} denote exporter–product–time, importer–product–time, and pair–product fixed effects.

Appendix 2

Table: EIA Effects on Agricultural Exports – Lagged Specification

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
All EIAs	0.061***	0.094***	0.062***	0.106***	0.061***	0.041**	0.062***	0.041**	0.063***	0.040*
	(0.019)	(0.022)	(0.019)	(0.023)	(0.019)	(0.021)	(0.023)	(0.021)	(0.024)	(0.021)
All EIAs$_{t-3}$	0.087***		0.066***		0.069***	0.035		0.045**		0.045**
	(0.020)		(0.017)		(0.017)	(0.022)		(0.018)		(0.018)
All EIAs$_{t-6}$		0.065***	0.037*		0.012		0.005	-0.014		-0.014
		(0.022)	(0.021)		(0.021)		(0.022)	(0.021)		(0.019)
All EIAs$_{t-9}$				0.068***	0.050**				0.000	0.001
				(0.021)	(0.020)				(0.024)	(0.024)
All EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$						0.189***	0.255***	0.197***	0.308***	0.194***
						(0.031)	(0.032)	(0.031)	(0.032)	(0.031)
All EIAs$_{t-3}$ $\times \mathbf{1}_{k \in EX_{ij}^{10}}$						0.228***		0.102***		0.117***
						(0.031)		(0.027)		(0.028)
All EIAs$_{t-6}$ $\times \mathbf{1}_{k \in EX_{ij}^{10}}$							0.224***	0.185***		0.089**
							(0.033)	(0.035)		(0.036)
All EIAs$_{t-9}$ $\times \mathbf{1}_{k \in EX_{ij}^{10}}$									0.222***	0.157***
									(0.031)	(0.032)
WTO	0.132	0.133	0.131	0.134	0.129	0.128	0.129	0.125	0.132	0.125
	(0.091)	(0.091)	(0.091)	(0.091)	(0.091)	(0.091)	(0.091)	(0.091)	(0.091)	(0.091)
Cumulative EIA effect (main + lags)										
All EIAs	0.148***	0.159***	0.164***	0.174***	0.191***	0.076**	0.067**	0.071**	0.063*	0.073*
	(0.028)	(0.032)	(0.033)	(0.033)	(0.036)	(0.030)	(0.034)	(0.035)	(0.037)	(0.040)
All EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$						0.417***	0.480***	0.484***	0.531***	0.557***
						(0.037)	(0.041)	(0.041)	(0.044)	(0.046)
N = 19,293,038 Pseudo $R^2 = 0.972$ FEs = ikt, jkt, ijk										

Appendix 3

$$Y_{ijkt} = \exp\Big(\beta_1^{xm}\left(EIA_{ijt} \times D_{xm}\right) + \beta_2^{xm}\left(EIA_{ijt} \times D_{xm} \times 1_{k \in EX_{ij}^{10}}\right) + \beta_3\, WTO_{ijt} + \mu_{ikt} + \gamma_{jkt} + \lambda_{ijk}\Big)\varepsilon_{ijkt} \tag{4}$$

Notes: x and m represent the income classification of the exporter and importer, respectively, such that x takes the value n (developed) or s (developing), and m likewise.

Appendix 4

Table: EIA Effects on Agricultural Exports – Lagged Specification

	(1) north-to-north	(2) north-to-south	(3) south-to-north
All EIAs	0.019	-0.282***	0.003
	(0.032)	(0.036)	(0.019)
All EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$	0.340***	0.765***	0.179***
	(0.040)	(0.044)	(0.042)
WTO	1.320**	1.520**	0.586***
	(0.563)	(0.621)	(0.221)
N	4,531,183	4,601,737	3,467,569
Pseudo R^2	0.979	0.979	0.982
FEs	ikt, jkt, ijk	ikt, jkt, ijk	ikt, jkt, ijk

Appendix 5

$$
\begin{aligned}
Y_{ijkt} = \exp\Big(&\beta_1\, EIA_{ijt} + \sum_p \beta_{2p}\,(EIA_{ijt} \times Provision^p_{ijt}) \\
&+ \sum_p \beta_{3p}\,(EIA_{ijt} \times Provision^p_{ijt} \times 1_{k \in EX^{10}_{ij}}) \\
&+ \beta_4\, WTO_{ijt} + \mu_{ikt} + \gamma_{jkt} + \lambda_{ijk}\Big)\varepsilon_{ijkt}
\end{aligned} \tag{5}
$$

Notes: The interaction $EIA_{ijt} \times Provision^p_{ijt}$ measures the intensive margin effect of each provision p, and the triple interaction $EIA_{ijt} \times Provision^p_{ijt} \times 1_{k \in EX^{10}_{ij}}$ isolates the extensive margin effect, capturing the additional impact of each provision on extensive margin goods.

Appendix 6

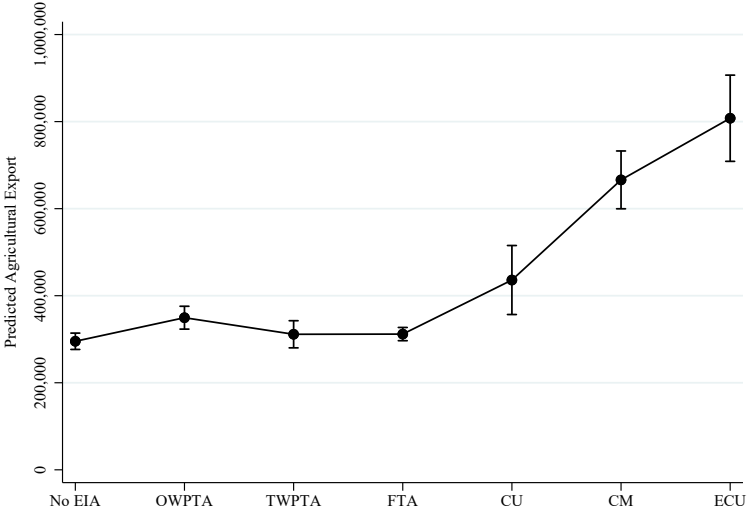
Table: Variance inflation factor test

Variable	VIF	1/VIF
All EIAs $\times$ Services	5.81	0.1721
All EIAs $\times$ Investments $\times \mathbf{1}_{k \in EX_{ij}^{10}}$	5.72	0.1748
All EIAs $\times$ Competition	5.66	0.1765
All EIAs $\times$ Standards $\times \mathbf{1}_{k \in EX_{ij}^{10}}$	5.61	0.1781
All EIAs $\times$ IPRs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$	5.00	0.2000
All EIAs $\times$ Investments	4.69	0.2133
All EIAs $\times$ Procurement $\times \mathbf{1}_{k \in EX_{ij}^{10}}$	4.53	0.2209
All EIAs $\times$ Services $\times \mathbf{1}_{k \in EX_{ij}^{10}}$	4.48	0.2230
All EIAs $\times$ Standards	4.22	0.2369
All EIAs $\times$ Competition $\times \mathbf{1}_{k \in EX_{ij}^{10}}$	4.11	0.2433
All EIAs $\times$ IPRs	2.92	0.3429
All EIAs $\times$ Procurement	2.82	0.3552
All EIAs	1.29	0.7742
WTO	1.04	0.9589
Mean VIF	4.14	

Appendix 7

	(1)	(2)
All EIAs	286,028***	
	(39,177)	
Shallow EIAs		158,858***
		(39,598)
Deep EIAs		653,439***
		(89,471)
WTO	238,969***	192,744***
	(22,473)	(22,739)
	N = 30,035,279	

Appendix 8

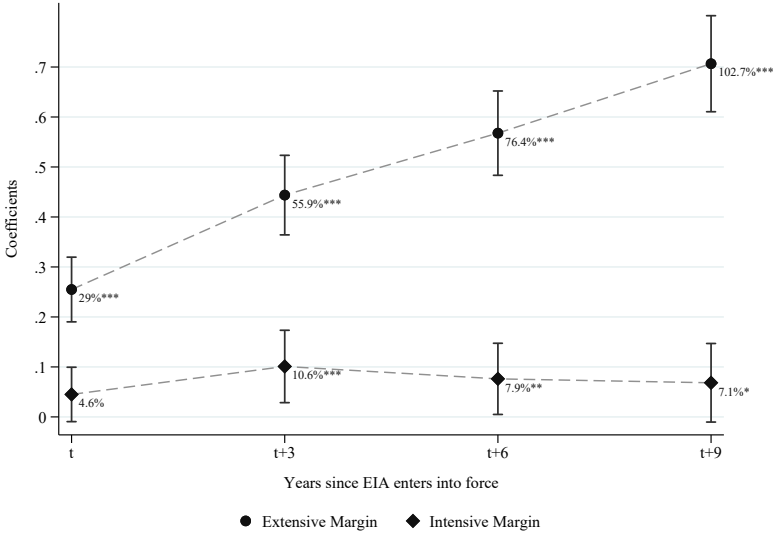

Predicted Agricultural Export
0
200,000
400,000
600,000
800,000
1,000,000
No EIA
OWPTA
TWPTA
FTA
CU
CM
ECU

Appendix 9

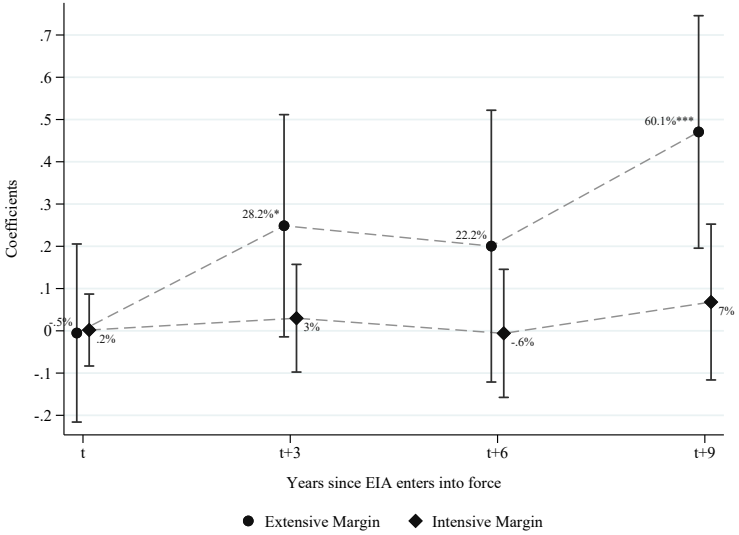
Table: EIA Effects on Raw and Processed Agricultural Exports – Lagged Specification

	Panel A. Processed Food Trade					Panel B. Raw Agricultural Trade				
	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
All EIAs	0.046*	0.073**	0.046*	0.071**	0.045	0.005	0.015	0.005	0.007	0.002
	(0.028)	(0.031)	(0.028)	(0.031)	(0.028)	(0.043)	(0.048)	(0.044)	(0.053)	(0.043)
All EIAs$_{t-3}$	0.037*		0.056***		0.056***	0.022		0.021		0.028
	(0.021)		(0.018)		(0.018)	(0.057)		(0.050)		(0.050)
All EIAs$_{t-6}$		-0.008	-0.030		-0.025		0.012	0.002		-0.036
		(0.026)	(0.027)		(0.022)		(0.051)	(0.045)		(0.044)
All EIAs$_{t-9}$				-0.014	-0.008				0.061	0.074
				(0.029)	(0.026)				(0.057)	(0.059)
All EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$	0.200***	0.280***	0.210***	0.359***	0.210***	0.004	0.113	0.007	0.150	-0.007
	(0.041)	(0.038)	(0.040)	(0.037)	(0.040)	(0.103)	(0.103)	(0.104)	(0.107)	(0.105)
All EIAs$_{t-3}$ $\times \mathbf{1}_{k \in EX_{ij}^{10}}$	0.288***		0.121***		0.133***	0.275***		0.193		0.226
	(0.033)		(0.034)		(0.034)	(0.102)		(0.135)		(0.140)
All EIAs$_{t-6}$ $\times \mathbf{1}_{k \in EX_{ij}^{10}}$		0.287***	0.240***		0.149***		0.199*	0.118		-0.012
		(0.031)	(0.033)		(0.027)		(0.104)	(0.132)		(0.164)
All EIAs$_{t-9}$ $\times \mathbf{1}_{k \in EX_{ij}^{10}}$				0.249***	0.147***				0.225**	0.196*
				(0.035)	(0.032)				(0.088)	(0.114)
WTO	-0.085	-0.079	-0.087	-0.074	-0.088	0.374**	0.373**	0.373**	0.375**	0.374**
	(0.084)	(0.084)	(0.084)	(0.084)	(0.083)	(0.183)	(0.183)	(0.183)	(0.183)	(0.183)
	Cumulative EIA effect (main + lags)									
All EIAs	0.083**	0.065*	0.071**	0.056	0.068*	0.027	0.027	0.028	0.067	0.068
	(0.034)	(0.035)	(0.036)	(0.037)	(0.040)	(0.072)	(0.080)	(0.081)	(0.088)	(0.094)
All EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$	0.488***	0.567***	0.571***	0.609***	0.638***	0.279**	0.311**	0.318**	0.375***	0.403***
	(0.039)	(0.041)	(0.041)	(0.045)	(0.046)	(0.120)	(0.135)	(0.135)	(0.133)	(0.139)
N	11,363,060	11,363,060	11,363,060	11,363,060	11,363,060	1,755,673	1,755,673	1,755,673	1,755,673	1,755,673
Pseudo R^2	0.079	0.079	0.079	0.079	0.079	0.079	0.079	0.079	0.079	0.079

Appendix 10


Coefficients
.7
.6
.5
.4
.3
.2
.1
0
102.7%***
76.4%***
55.9%***
29%***
10.6%***
7.9%**
7.1%*
4.6%
t
t+3
t+6
t+9
Years since EIA enters into force
Extensive Margin
Intensive Margin

Appendix 11


.7
.6
.5
.4
.3
.2
.1
0
-.1
-.2
Coefficients
-.5%
.2%
28.2%*
3%
22.2%
-.6%
60.1%***
7%
t
t+3
t+6
t+9
Years since EIA enters into force
Extensive Margin
Intensive Margin

Appendix 12

Table: Further on margins

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
All EIAs	0.061**	0.007	0.072***	0.056***	0.064**	0.063**	0.102***	0.047***	0.078***	0.044***
	(0.025)	(0.034)	(0.012)	(0.013)	(0.026)	(0.025)	(0.017)	(0.016)	(0.012)	(0.013)
All EIAs $\times \mathbf{1}_{k \in EX^{10}_{ij}}$	0.360***	0.555***						0.379***		0.236***
	(0.033)	(0.056)						(0.024)		(0.022)
All EIAs $\times \mathbf{1}_{k \in EX^{5}_{ij}}$			0.398***							
			(0.024)							
All EIAs $\times \mathbf{1}_{k \in EX^{15}_{ij}}$				0.316***						
				(0.021)						
All EIAs $\times \mathbf{1}_{k \in EX^{10}_{ij(NEW)}}$					0.327***					
					(0.030)					
All EIAs $\times \mathbf{1}_{k \in EX^{10}_{ij(MAX)}}$						0.372***				
						(0.032)				
All EIAs $\times \mathbf{1}_{k \in EX^{10}_{ij(REL)}}$							0.029			
							(0.020)			
All EIAs $\times \mathbf{1}_{k \in EX^{10-40}_{ij}}$								0.075***		
								(0.024)		
All EIAs $\times \mathbf{1}_{k \in EX^{40-70}_{ij}}$								-0.021		
								(0.035)		
Trend$_t \times \mathbf{1}_{k \in EX^{10}_{ij(1996)}}$									0.043***	0.043***
									(0.001)	(0.001)
WTO	0.137	0.074	0.136**	0.136**	0.134	0.137	0.139**	0.137**	-0.282*	-0.287*
	(0.091)	(0.115)	(0.065)	(0.065)	(0.091)	(0.091)	(0.065)	(0.065)	(0.151)	(0.152)
N	19,293,038	21,977,417	19,293,038	19,293,038	19,293,038	19,293,038	19,293,038	12,926,281	12,893,754	19,293,038
Pseudo R^2	0.972	0.932	0.971	0.971	0.972	0.972	0.971	0.976	0.975	0.971
FEs	ikt, jkt, ijk	it, jt, ijk	ikt, jkt, ijk	ikt, jkt, ijk	ikt, jkt, ijk	ikt, jkt, ijk	ikt, jkt, ijk	ikt, jkt, ijk	ikt, jkt, ijk	ikt, jkt, ijk

Appendix 13

Table: Aggregated at commodity group level

	(1)	(2)	(3)	(4)	(5)
All EIAs	0.116***		0.081***	0.062***	
	(0.023)		(0.024)	(0.021)	
All EIAs$_{t-3}$				0.032	
				(0.022)	
Shallow EIAs		0.125***			0.091***
		(0.023)			(0.024)
Deep EIAs		0.767***			0.492***
		(0.066)			(0.065)
All EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$			0.265***	0.070*	
			(0.040)	(0.036)	
All EIAs$_{t-3}$ $\times \mathbf{1}_{k \in EX_{ij}^{10}}$				0.253***	
				(0.035)	
Shallow EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$					0.260***
					(0.039)
Deep EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$					1.019***
					(0.059)
WTO	0.086	0.106	0.083	0.073	0.104
	(0.083)	(0.082)	(0.083)	(0.083)	(0.082)

N = 5,050,563 Pseudo R^2 = 0.972 FEs = ikt, jkt, ijk

Appendix 14

Table: Aggregated at commodity group level - Full Sample

	(1)	(2)	(3)	(4)	(5)
All EIAs	0.103***		0.061***	0.053***	
	(0.021)		(0.023)	(0.019)	
All EIAs$_{t-3}$				0.016	
				(0.020)	
Shallow EIAs		0.113***			0.072***
		(0.021)			(0.023)
Deep EIAs		0.734***			0.457***
		(0.063)			(0.063)
All EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$			0.309***	0.113***	
			(0.038)	(0.032)	
All EIAs$_{t-3}$ $\times \mathbf{1}_{k \in EX_{ij}^{10}}$				0.260***	
				(0.031)	
Shallow EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$					0.302***
					(0.037)
Deep EIAs $\times \mathbf{1}_{k \in EX_{ij}^{10}}$					1.025***
					(0.056)
WTO	0.089	0.108	0.085	0.077	0.104
	(0.075)	(0.075)	(0.075)	(0.075)	(0.075)

N = 17,718,758 Pseudo $R^2 = 0.972$ FEs = ikt, jkt, ijk

Appendix 15

Table: Aggregated data

	(1)	(2)	(3)	(4)	(5)	(6)
All EIAs	0.087***				0.029	
	(0.032)				(0.029)	
All EIAs$_{t-3}$					0.087***	
					(0.030)	
Shallow EIAs		0.094***		0.096***		0.031
		(0.032)		(0.032)		(0.029)
Shallow EIAs$_{t-3}$						0.094***
						(0.030)
Deep EIAs		0.581***	0.551***			0.397***
		(0.076)	(0.077)			(0.077)
Deep EIAs$_{t-3}$						0.255***
						(0.062)
One-way PTAs			0.146***			
			(0.041)			
Two-way PTAs			0.081			
			(0.053)			
FTAs			0.064*			
			(0.036)			
Custom Unions				0.371***		
				(0.101)		
Common Markets				0.744***		
				(0.079)		
Economic Unions				0.929***		
				(0.098)		
WTO	0.204*	0.232*	0.231*	0.278**	0.198	0.238**
	(0.122)	(0.121)	(0.121)	(0.122)	(0.122)	(0.121)

Appendix 16

Table: Aggregation bias

	1	2
All EIAs	0.084***	0.117***
	(0.031)	(0.023)
WTO	0.075	0.138
	(0.114)	(0.091))
Fixed Effects	it, jt, ij	ikt, jkt,ijk
N	29,584,527	19,293,038